Operational Weekly Dengue Forecasting in Bangladesh: Hierarchy-Aware Panel Backtesting Shows Strong Baseline Dominance Under Recent-Window Stress Testing

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Abstract

Background: Dengue has become a recurrent national public health threat in Bangladesh, yet many published dengue forecasting studies still emphasize retrospective fit, climate-heavy predictor sets, or limited validation designs that do not map cleanly to operational deployment [1–3]. We therefore reframed the task as weekly short-horizon forecasting of dengue case counts by geography under real surveillance constraints.

Methods: We assembled a weekly panel of 9 Bangladesh geographies from 2021-12-27 to 2025-09-15 and repaired an overlapping Dhaka hierarchy by replacing the original division-level series with a non-overlapping out-of-metro residual geography. We first performed broad exploratory diagnostics, including stationarity tests, seasonal decomposition, autocorrelation analysis, rolling moments, zero-inflation checks, overdispersion summaries, hierarchy checks, and climate lead-lag analysis. We then evaluated two direct multi-horizon forecasting tracks: an operational 1–4 week track using case-history and calendar predictors only, and an experimental 5–12 week track that additionally used lagged climate features. Because the latest 2025 block was used for diagnostic refinement, the evidence hierarchy was refrozen into *development*, *selection*, and *diagnostic* windows rather than a single untouched final holdout. Macro-averaged mean absolute error (MAE) across geography and horizon was the primary metric.

Results: The panel was highly heterogeneous but strongly structured: zero-week proportions ranged from 3.1% to 55.9%, variance-to-mean ratios from 61.4 to 2797.1, lag-1 autocorrelation from 0.953 to 0.980, and lag-52 autocorrelation from -0.078 to 0.219. Climate coverage ended 11 weeks before the latest case week for every geography. In operational development folds, a conservative persistence-anchored ensemble was best (MAE 153.448; RMSE 196.133). In the fixed operational selection window, several challengers outperformed persistence, with `panel_hist_gbm` best (MAE 18.429; RMSE 36.923) versus persistence (MAE 33.993; RMSE 61.040). However, in the most recent diagnostic window, persistence again dominated (MAE 92.571; RMSE 125.059) and all challengers deteriorated materially. In the experimental track, `ar_tweedie_global` was best on selection (MAE 35.423), whereas persistence was best on the diagnostic window (MAE 183.073).

Interpretation: The most defensible result is not that one challenger model has already replaced persistence, but that challenger gains are window-dependent and do not yet survive recent-window stress testing. In this data environment, operational dengue forecasting in Bangladesh is governed primarily by short-run case momentum, hierarchy consistency, and robustness under regime shift. Until new post-tuning data are available for a truly untouched final evaluation block, persistence remains the appropriate primary operational forecast.

1 Introduction

Dengue is one of the most important mosquito-borne viral diseases globally, with increasing burden over recent decades and substantial impact in tropical and subtropical countries [1, 4]. Bangladesh has experienced a marked escalation in dengue burden, with larger outbreaks, broader geographic spread, and increasing operational pressure on hospitals and public health services [5]. The 2023 outbreak made the forecasting problem especially urgent, as transmission expanded beyond Dhaka and accelerated under conditions that challenged routine preparedness assumptions [3].

Short-horizon forecasting could support bed planning, resource allocation, and situational awareness. However, dengue forecasting is difficult in operational settings because the signal is shaped simultaneously by epidemic momentum, spatial heterogeneity, intermittent zeros in lower-burden areas, incomplete exogenous data streams, and abrupt regime shifts. A systematic review of dengue prediction models found heavy reliance on climate predictors, inconsistent validation, and limited evidence of real-world generalization [2]. Those limitations matter most when the goal is deployment rather than retrospective explanation.

Bangladesh-specific forecasting studies have contributed useful evidence, including time-series and machine learning analyses at national or division-level resolution [6, 7]. But an operational gap remains. We were not interested in identifying the most flexible model in the abstract. Instead, we asked a more practical question: for weekly dengue surveillance in Bangladesh, what forecasting formulation is most trustworthy for 1–4 week operational use, and what does a transparent, hierarchy-aware, rolling-origin comparison actually show?

That question is useful whether the answer is positive or negative. If a challenger model provides stable gains, it supports escalation. If strong simple baselines remain dominant after careful diagnostics and realistic backtesting, that is still actionable evidence for both operations and future research. The present study therefore emphasizes honest task framing, data diagnostics, evaluation design, and robustness of conclusions over headline model novelty alone.

2 Related Work and Study Contribution

Three strands of prior work shaped this study. First, dengue forecasting reviews emphasize the field's dependence on climate-rich predictors and its inconsistent validation practices [2]. Second, Bangladesh-specific studies show that both classical time-series methods and machine learning approaches can capture aspects of dengue dynamics, particularly at monthly resolution [6, 7]. Third, hierarchical forecasting and outbreak-forecast evaluation studies demonstrate that forecast design, aggregation logic, and evaluation windows can matter as much as model class [8–10].

Against that background, our study makes five contributions.

1. It defines the task explicitly as weekly, multi-geography, direct multi-horizon forecasting of dengue case counts.

2. It repairs an overlapping Dhaka hierarchy before modeling so that bottom-level series are coherent.

3. It documents a broad exploratory diagnostic workflow rather than treating EDA as an unpublished preprocessing step.

4. It separates operational and experimental modeling tracks according to what data are truly usable in deployment.

5. It shows that model ranking is highly window-dependent: challengers can beat persistence on a fixed selection window, yet persistence remains dominant on the most recent diagnostic stress test.

That last contribution is especially important for an applied forecasting paper. A well-supported robustness result is more useful than an overfit claim of superiority. The present manuscript therefore focuses on what the data actually support, including where the evidence remains provisional.

3 Data, Problem Statement, and Scope

3.1 Data Sources

Weekly dengue case counts were assembled from Bangladesh surveillance releases curated in the project pipeline. Climate covariates were aggregated to weekly geography-level series from the configured ERA5/IMERG workflow. The case panel spans 2021-12-27 to 2025-09-15. Climate covariates span 2021-12-20 to 2025-06-30, leaving an 11-week coverage gap relative to the most recent case observations.

Each geography contributed 195 weekly case observations with no missing weeks after weekly aggregation. The original geography set contained an overlap in which the Dhaka division series included Dhaka Metro. Because coherent bottom-level modeling requires disjoint units, we replaced the overlapping division series with an out-of-metro residual geography (`DHA_DIV_OUT_METRO`). After this repair, the hierarchy closed exactly with zero residual error.

3.2 Problem Statement

The target variable was weekly dengue case counts at the `geo_id` level on the original count scale. We framed the primary task as short-horizon operational forecasting for 1–4 weeks ahead. A secondary experimental task covered 5–12 weeks ahead to test whether lagged climate information might help at longer leads. Forecasts were generated for each geography and horizon, then scored per geography and per horizon before macro-averaging. This design prevented Dhaka-dominant volumes from masking poor performance elsewhere.

3.3 Scope and Evaluation Philosophy

The study was designed for operational realism. That meant prioritizing point forecasting over probabilistic intervals, using only origin-available predictors in the operational track, and ranking models with macro-averaged MAE rather than pooled country-total error. It also meant being conservative about model promotion. Once the latest 2025 block had been used diagnostically to understand failure modes, it could no longer be presented as an untouched final holdout. We therefore refroze the evaluation protocol into evidence tiers rather than overstating certainty.

Table 1: Study design summary.

Item	Value
Forecast target	Weekly dengue case counts
Temporal grain	Weekly
Spatial grain	`geo_id`
Operational horizon	1 to 4 weeks ahead
Experimental horizon	5 to 12 weeks ahead
Primary KPI	Macro-averaged MAE across geography and horizon
Secondary KPI	Macro-averaged RMSE
Forecast strategy	Direct multi-horizon, not recursive
Operational covariates	Case-history and calendar features only
Experimental covariates	Case-history, calendar, and climate features
Hierarchy repair	`DHA_DIV` replaced by `DHA_DIV_OUT_METRO`
Evidence hierarchy	Development, selection, and diagnostic windows

Table 2: Refrozen evaluation protocol after diagnostic reuse of the latest 2025 block.

Tier	Fold(s)	Validation window	Purpose
Development	`D1--D4`	2023-01-01 to 2024-12-31 in rolling windows	Model building, debugging, and feature design
Selection	`S1`	2025-01-01 to 2025-06-30	Fixed pre-diagnostic challenger comparison
Diagnostic	`X1`	2025-07-01 to 2025-09-15	Stress testing on the latest available window; not a final untouched lockbox

4 Exploratory Data Analysis and Time-Series Diagnostics

We treated EDA as part of the scientific record rather than as invisible preprocessing. The diagnostic workflow included missing-week checks, combined ADF/KPSS stationarity screening, STL decomposition, autocorrelation functions, rolling means and rolling standard deviations, weekly seasonal profiles, zero-inflation summaries, overdispersion checks, cross-geography correlation, hierarchy checks, and climate lead-lag analysis.

The panel was heterogeneous in both scale and sparsity. Mean weekly cases ranged from 10.1 in Sylhet to 1003.5 in Dhaka Metro. Zero-week proportions ranged from 3.1% to 55.9%. Variance-to-mean ratios ranged from 61.4 to 2797.1, confirming strong overdispersion in every geography. Short-run dependence was very strong, with lag-1 autocorrelation from 0.953 to 0.980. By contrast, lag-52 autocorrelation ranged from -0.078 to 0.219, indicating that seasonality existed but was much weaker than recent momentum.

The stationarity question required nuance. All raw series satisfied the combined ADF/KPSS screening rule as stationarity candidates, eight of nine `log1p`-transformed series also passed, and all first-differenced series passed. Formally, the series therefore do not look like simple unit-root processes. Practically, however, rolling moments changed sharply through outbreak periods, especially during the 2023 wave. We therefore interpreted the data as outbreak-driven, heteroskedastic, and highly persistent rather than as smoothly weak-stationary in a modeling sense.

Seasonality was real but not dominant enough to drive operational forecasting on its own. Geography-specific peak weeks clustered broadly in the late monsoon and post-monsoon period. STL decomposition showed visible seasonal structure in several geographies, especially Dhaka-centered series and Khulna. Yet the seasonal naive benchmark performed poorly in backtesting. The key EDA implication was that seasonal features were appropriate, but same-week-last-year recurrence was too weak to serve as the primary operational signal.

Climate diagnostics revealed both promise and limitation. Dew point, temperature, humidity, rain, and soil moisture showed plausible lead-lag associations with future cases, often peaking around 8–10 weeks. But climate coverage ended 11 weeks before the latest case week in every geography. This was a pipeline constraint, not a geography-specific issue, and it directly motivated the separation between operational and experimental tracks.

Table 3: Condensed diagnostic summary of the weekly case panel after hierarchy repair.

Diagnostic	Observed range or result	Interpretation
Weekly mean cases by geography	10.1 to 1003.5	Strong scale imbalance; pooled metrics would be Dhaka-dominated
Zero-week proportion	3.1% to 55.9%	Mix of dense and sparse count processes
Variance-to-mean ratio	61.4 to 2797.1	Severe overdispersion beyond simple Poisson assumptions
Lag-1 autocorrelation	0.953 to 0.980	Short-run persistence is the dominant predictive signal
Lag-52 autocorrelation	−0.078 to 0.219	Annual repetition exists but is weak to modest
ADF/KPSS screen	9 of 9 raw series passed	No strong unit-root evidence, but this did not imply stable raw dynamics
Rolling moments	Large shifts during outbreak windows	Practical heteroskedasticity and regime dependence
Climate coverage gap	11 weeks for every geography	Weather could not be the required real-time production signal

5 Methods

5.1 Forecasting Strategy

We used direct multi-horizon forecasting rather than recursive forecasting [9]. Each horizon had its own prediction rule or fitted model. Forecasts at horizon 1 were not fed into horizon 2, horizon 2 forecasts were not fed into horizon 3, and so on. This choice avoided recursive error accumulation and made horizon-specific behavior easier to interpret in a short, noisy epidemic panel.

5.2 Dependent Variables

Two target formulations were used.

1. **Direct count models:** for models such as `ar_tweedie_global` and `panel_hist_gbm`, the

dependent variable was the future observed count,

$$y_{t+h},$$

where t is the forecast origin and h is the forecast horizon.

2. **Residual correction models:** for the residual boosting, random-forest, and elastic-net models, the dependent variable was the deviation from persistence,

$$y_{t+h} - y_t.$$

Final forecasts were reconstructed as current observed cases plus the predicted residual correction.

5.3 Independent Variables

The operational track used only variables available at the forecast origin:

- current weekly cases,
- case lags at 1, 2, 3, 4, 8, 13, 26, and 52 weeks,
- rolling case summaries over 2, 4, and 8 weeks,
- `log1p(cases)`,
- calendar features including week-of-year, sine/cosine seasonal terms, and a monsoon indicator,
- geography identity for pooled global models.

The experimental track added lagged and rolled climate covariates for rain, near-surface temperature, humidity, dew point, and shallow soil moisture. Weather was intentionally excluded from the operational core because of the 11-week coverage gap at the end of the series.

5.4 Model Ladder

The model ladder was chosen to answer a practical benchmarking question rather than to maximize algorithmic variety.

- **Persistence:** future cases equal the latest observed count.
- **Seasonal naive$_{52}$:** future cases equal the count observed 52 weeks earlier.
- **AR Tweedie global:** a pooled autoregressive count model using lag and calendar features.
- **Panel HistGBM:** a pooled nonlinear model fit directly on future counts.
- **Residual HistGBM:** a persistence-anchored boosting model fit on residual departures from the naive baseline.
- **Residual random forest:** a second persistence-anchored residual model.

- **Residual elastic net global:** a pooled regularized linear residual model.
- **Conservative stack:** a nonnegative horizon-specific ensemble anchored to persistence and weighted using development-fold MAE.

We deliberately did not make deep learning part of the primary operational ladder. With only 195 weekly observations per series and 9 series overall, high-capacity sequence models are more likely to overfit than to provide stable operational gains at this stage.

5.5 Operational Safeguards and Diagnostic Refinement

Trainable models were subject to stability guards. Fits that produced implausibly explosive forecasts relative to the observed training range were reverted to persistence. After the initial challenger pass, we also investigated recent-window failure modes and added low-origin guardrails so that some models would not overpredict rebounds immediately after zero or near-zero origins. Those diagnostic refinements improved specific error regimes, but because they used the latest 2025 data diagnostically, that window could no longer be treated as a final untouched holdout. This is why the manuscript reports a refrozen evidence hierarchy rather than a single definitive post-tuning test.

5.6 Metrics

The primary outcome was macro-averaged MAE across geography and horizon. RMSE was secondary. Metrics were computed per geography and per horizon before averaging. Pooled totals were retained for descriptive use only and were not used for model promotion.

6 Results

6.1 Development-Fold Behavior

Across the four development folds, the operational model ranking favored the conservative stack overall (MAE 153.448; RMSE 196.133), followed by persistence (MAE 167.212; RMSE 216.485), residual random forest (MAE 170.061), and AR Tweedie global (MAE 175.108). This indicated that persistence-anchored residual learning was a plausible improvement path during model development.

At the fold level, however, no single challenger dominated all regimes. The development heatmap shows that some models won calm windows while others were stronger under outbreak-heavy conditions. This was the first warning that global average performance could mask substantial temporal instability.

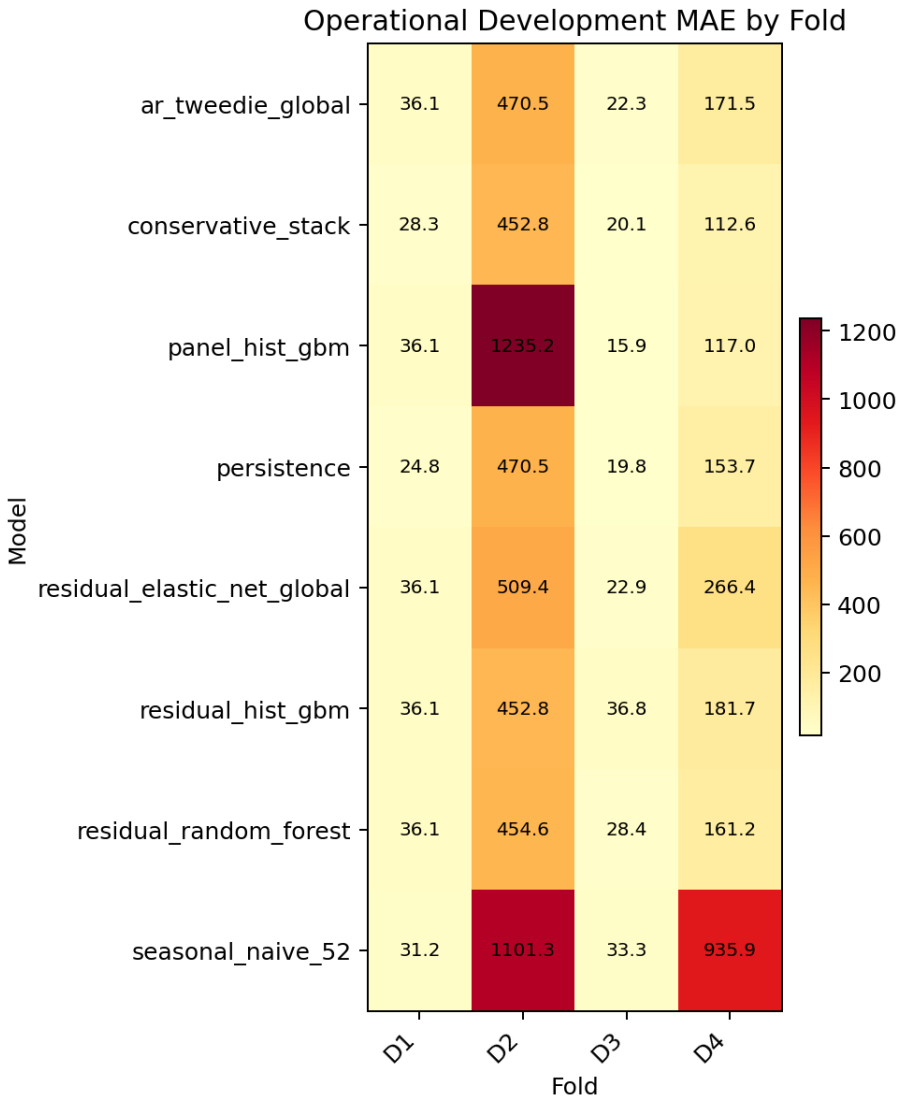

Figure 1: Operational development-fold MAE. Model ranking changes across windows, indicating that apparent gains during model building are regime-dependent rather than universal.

6.2 Operational Selection Window

The fixed operational selection window produced the most encouraging challenger results. The best model was panel HistGBM (MAE 18.429; RMSE 36.923), followed closely by the residual elastic-net model (MAE 18.829; RMSE 35.618) and the global AR Tweedie model (MAE 19.901; RMSE 35.575). Persistence ranked seventh on MAE (33.993) in this window. The selection result therefore showed that the dataset does contain learnable structure beyond pure persistence.

Table 4: Operational leaderboard on the fixed selection window (2025-01-01 to 2025-06-30).

Model	MAE	RMSE
panel_hist_gbm	18.429	36.923
residual_elastic_net_global	18.829	35.618
ar_tweedie_global	19.901	35.575
conservative_stack	23.887	43.383
residual_random_forest	25.216	44.040
residual_hist_gbm	31.902	59.994
persistence	33.993	61.040
seasonal_naive_52	39.410	77.366

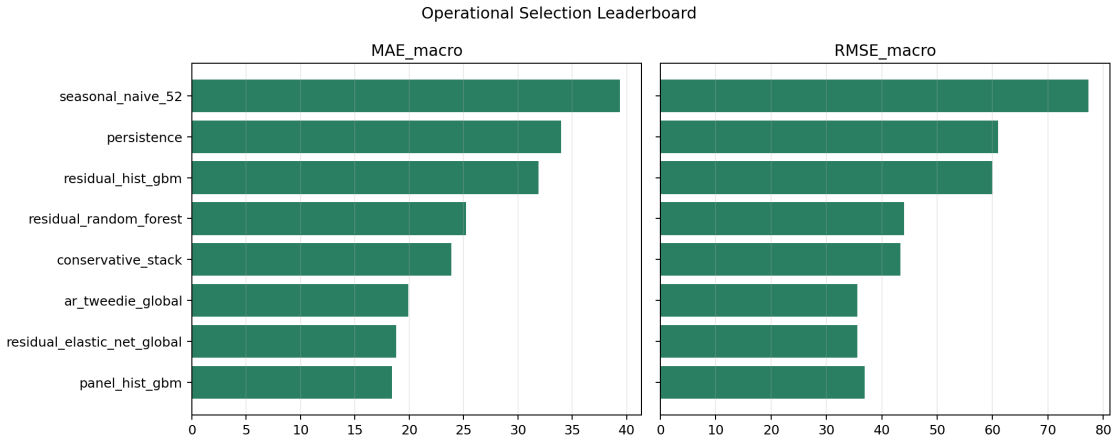

Figure 2: Operational selection leaderboard. Several challengers outperform persistence convincingly in the fixed pre-diagnostic comparison window.

Selection-window horizon winners also varied by lead time: residual random forest was best at horizon 1, panel HistGBM at horizons 2 and 4, and residual elastic net at horizon 3. This supported the idea that different model families captured different local regimes within the same window.

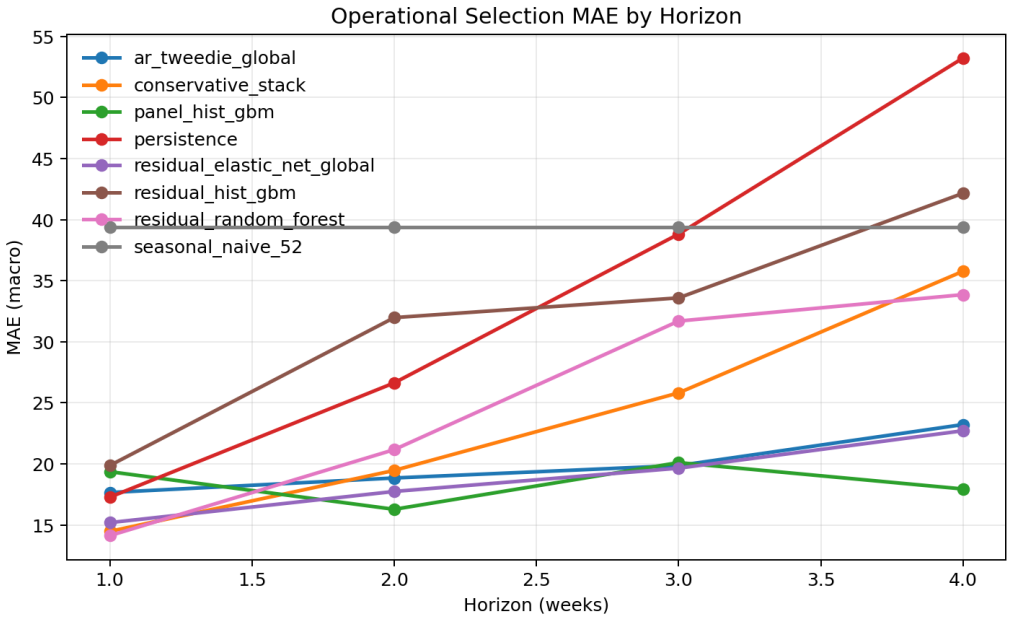

Figure 3: Operational selection-window MAE by horizon. Challenger wins were not confined to a single lead time.

6.3 Operational Diagnostic Window

The story changed sharply on the most recent diagnostic window. Here persistence regained clear dominance, with MAE 92.571 and RMSE 125.059. The next-best model was the conservative stack at MAE 177.161, followed by seasonal naive$_{52}$ at 226.465. All other challengers were materially worse. Persistence also won every diagnostic operational horizon, from 1 week ahead (MAE 43.737) through 4 weeks ahead (MAE 146.424).

Table 5: Operational leaderboard on the diagnostic window (2025-07-01 to 2025-09-15).

Model	MAE	RMSE
`persistence`	92.571	125.059
`conservative_stack`	177.161	209.539
`seasonal_naive_52`	226.465	283.906
`residual_random_forest`	238.695	299.302
`residual_elastic_net_global`	247.707	286.439
`panel_hist_gbm`	280.907	349.719
`ar_tweedie_global`	288.643	328.117
`residual_hist_gbm`	303.688	404.871

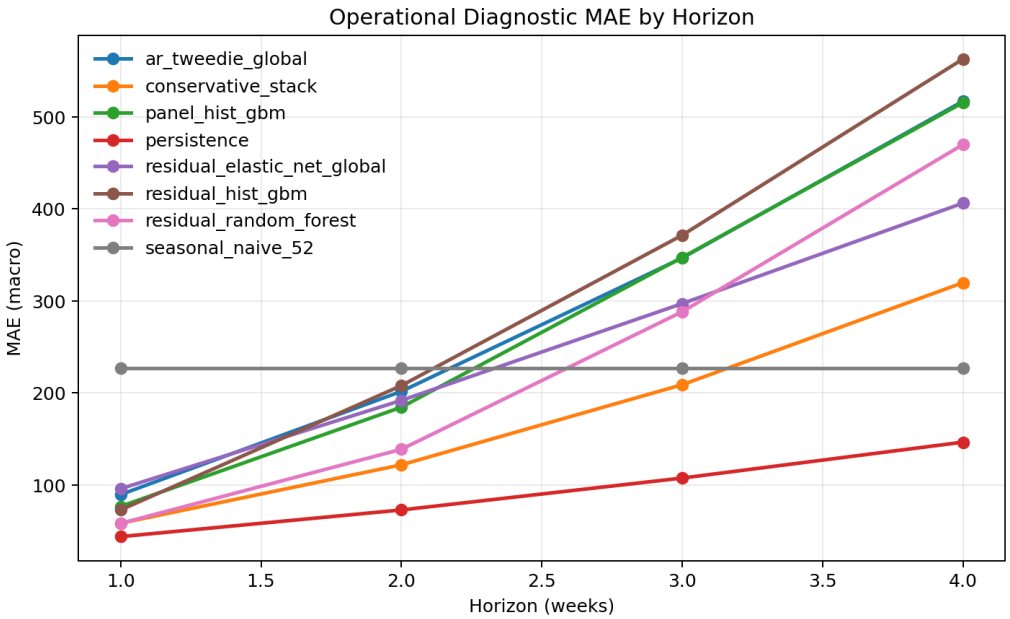

Figure 4: Operational diagnostic-window MAE by horizon. Persistence is the best model at every operational lead time in the latest stress-test window.

This disagreement between selection and diagnostic evidence is the central empirical result of the study. If we had stopped at the selection window, we could have claimed challenger superiority. But once the recent diagnostic window is included honestly, the more defensible conclusion is that challenger gains are not yet robust enough for operational promotion.

6.4 Experimental Track

The experimental 5–12 week track told a similar story. On the fixed selection window, the global AR Tweedie model was best (MAE 35.423; RMSE 61.112), narrowly ahead of panel HistGBM (MAE 36.231), while persistence performed poorly (MAE 178.336). On the diagnostic window, however, persistence was again best (MAE 183.073; RMSE 199.499), ahead of seasonal naive$_{52}$ (MAE 195.636), with climate-augmented challengers degrading sharply.

Table 6: Experimental track leaderboards on the selection and diagnostic windows.

Window	Model	MAE	RMSE
Selection	`ar_tweedie_global`	35.423	61.112
Selection	`panel_hist_gbm`	36.231	65.192
Selection	`seasonal_naive_52`	39.410	77.366
Selection	`persistence`	178.336	300.890
Diagnostic	`persistence`	183.073	199.499
Diagnostic	`seasonal_naive_52`	195.636	232.594
Diagnostic	`ar_tweedie_global`	537.848	680.006
Diagnostic	`panel_hist_gbm`	673.757	861.092

This does not imply that climate is irrelevant to dengue transmission. Rather, it shows that the current climate-augmented forecasting setup does not produce a robust operational advantage under the available coverage window and sample size.

6.5 Interpretation of Model Behavior

The model ladder revealed distinct behavioral patterns. The conservative stack was the strongest model during development because it stayed close to persistence while borrowing controlled signal from challenger models. Panel HistGBM and elastic-net residual correction were strong on the selection window, which suggests the presence of meaningful nonlinear and linearizable deviations from persistence. Yet those gains did not survive the latest diagnostic window. This is exactly the kind of instability that a top-line leaderboard can conceal.

The failure-mode analysis also showed that trainable models were vulnerable to rebound overprediction after low-count origins and to degradation under regime shifts. Guardrails reduced those specific failure modes, but they did not overturn the diagnostic dominance of persistence. The central issue is therefore not that challengers have no signal. It is that the signal they capture is not yet stable enough across temporal regimes.

6.6 Current Operational Forecast

Because there is no longer a truly untouched post-tuning holdout and persistence remained dominant on the diagnostic window, the live forecasting policy remained conservative. The system uses the selection window for challenger ranking, but it does not auto-promote a challenger without a new untouched final block. As a result, the current primary model for the origin week 2025-09-15 is persistence.

At the country-total level, the persistence forecast is 0.0 for horizons 1–4, reflecting zero observed bottom-level counts at the latest origin. Challengers allow small to moderate rebounds, ranging from 0.8 to 27.9 for the conservative stack and panel HistGBM, while seasonal naive_{52} produces implausibly large values anchored to the previous year's outbreak season.

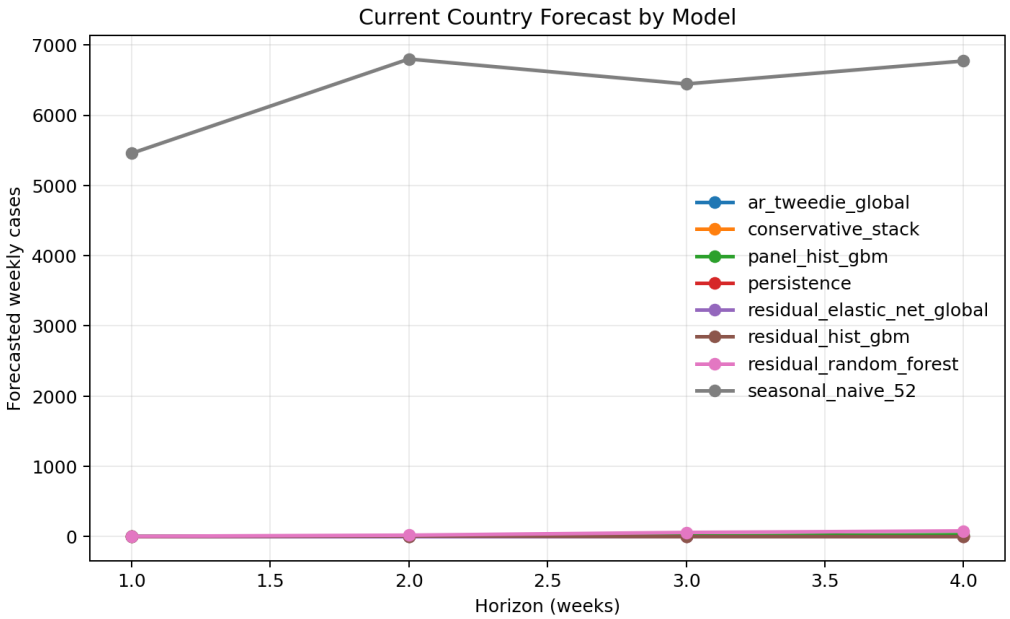

Figure 5: Illustrative current country-total forecasts from the refrozen protocol. Persistence remains the promoted primary model because challenger ranking is based on the selection window only and there is no new untouched final block for promotion.

7 Discussion

This study set out to answer an operational forecasting question rather than a model-novelty question. The answer is more nuanced than a simple leaderboard win. The data clearly contain structure beyond persistence: multiple challengers improved materially on the fixed selection window, and the development folds supported persistence-anchored residual learning. However, the most recent diagnostic window still strongly favored persistence. That means the strongest scientific claim is about robustness, not about premature model replacement.

Several insights follow from this result. First, the dominant short-horizon signal in the current Bangladesh panel is recent case momentum. That conclusion is supported jointly by the EDA, the poor seasonal-naive performance, and the continued diagnostic dominance of persistence. Second, the presence of plausible climate associations does not automatically translate into operational forecasting gains when the climate data stream is incomplete at the point of deployment. Third, model development should focus less on free-running complexity and more on controlled deviation from persistence, regime awareness, and sparse-series behavior.

The divergence between selection and diagnostic windows also has methodological implications. In outbreak forecasting, model ranking is often window-dependent [10]. Our results show this clearly. Had we reported only the selection window, the paper would have favored panel HistGBM or elastic-net residual correction. By preserving the diagnostic record and refreezing the protocol after diagnostic reuse, we instead show how fragile those gains are under the latest available conditions. That makes the paper stronger, not weaker.

For public health practice, the immediate implication is conservative but actionable: a simple, transparent persistence baseline remains the safest current operational model for 1–4 week weekly

dengue forecasting in Bangladesh. Challengers should continue to be developed, but they should not replace persistence until they can win on a newly frozen untouched future block.

8 Limitations

This study has several limitations. First, the panel is still short for high-capacity multi-series forecasting, with only 195 weekly observations per geography. Second, a small number of large outbreak windows contribute heavily to the overall signal. Third, the climate series end 11 weeks before the latest case week, which sharply limits operational weather use. Fourth, the present analysis focused on point forecasting; calibrated uncertainty quantification remains future work. Fifth, once the latest 2025 block had been used for post-hoc diagnostic refinement, no fully untouched final holdout remained within the available data. Consequently, challenger gains in this paper should be interpreted as promising but not yet final. Sixth, the study is based on one national surveillance system and has not yet been externally validated in a separate country or institutional deployment context.

9 Conclusion

We reframed dengue forecasting in Bangladesh as a hierarchy-aware, weekly, multi-geography count forecasting problem and evaluated a transparent ladder of baselines and challenger models under rolling-origin backtesting. The data exhibit strong persistence, substantial overdispersion, mixed sparsity, weak-to-moderate annual seasonality, and incomplete recent climate coverage. Under those conditions, several challengers outperform persistence on a fixed selection window, but persistence remains dominant on the most recent diagnostic stress test. The most defensible operational conclusion is therefore conservative: persistence should remain the primary model until new post-tuning data permit a truly untouched final evaluation block. The most promising next research steps are regime-aware residual correction, sparse-series hurdle formulations, improved real-time covariate availability, and future evaluation under a newly frozen holdout window.

Data Availability

Processed datasets, artifact bundles, figures, and manuscript assets used in this study are archived in the project repository. Access to upstream raw surveillance and environmental data should remain subject to the original providers' terms and reporting policies.

Code Availability

All code used to construct the hierarchy-fixed panel, run EDA, execute rolling-origin backtests, generate current forecasts, and build manuscript figures and tables is available in the accompanying project repository.

Competing Interests

The authors declare that they have no competing interests.

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